Nervous System

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MPhil – Anatomy
April 15th, 2014

Nervous System

Latin : nervus means nerve

Nerve literally means Unduly excitable

Excitable cells in the brain controls and coordinates other parts of the body

Excitable cells or nerve cells sit in brain and sends extensions to all parts of the body

Nerve cells are called **Neurons**; situated in brain or center

Extensions of neurons are called **Axon** and are away from center i.e. brain

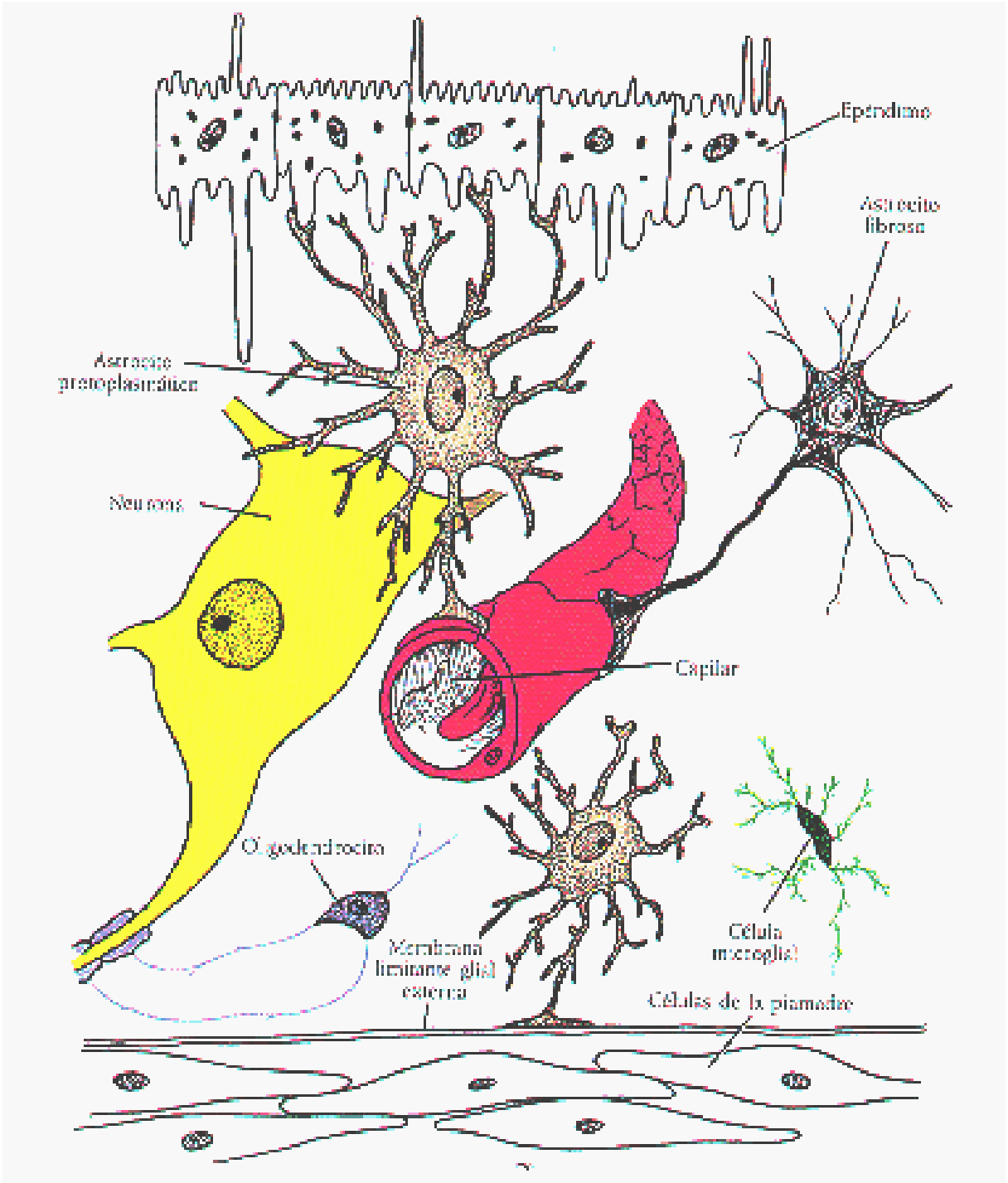
Nervous tissue is a specialized connective tissue

Neurons are the basic cells and the support tissue is called **Glia** or **Neuroglia**

Neuroglia are non- conducting cells

Neuroglia

Astrocytes –	Support & Nutrition
Microglia –	Immune cell i.e. Monocyte
Schwann cells –	Myelin sheath in PNS
Oligodendrocytes –	Myelin sheath in CNS
Ependymal cells –	CSF formation


Ependimo
Astrocito fibroso
Astrocito protoplasmático
Neurona
Capilar
Oligodendrocito
Membrana limitante glial externa
Célula microglial
Células de la piamadre

Receptor Terminal

- *Soma / Perikaryon*
- *Dendrites—Arborization*

Conductile Portion

- *Axon hillock*
- *Axon --- 1 or 2*
- *Myelin--- oligodendrocytes*
- *Nodes of Ranvier*

Synaptic Terminal

- *Arborization*
- *Synaptic ends---Terminal boutons*

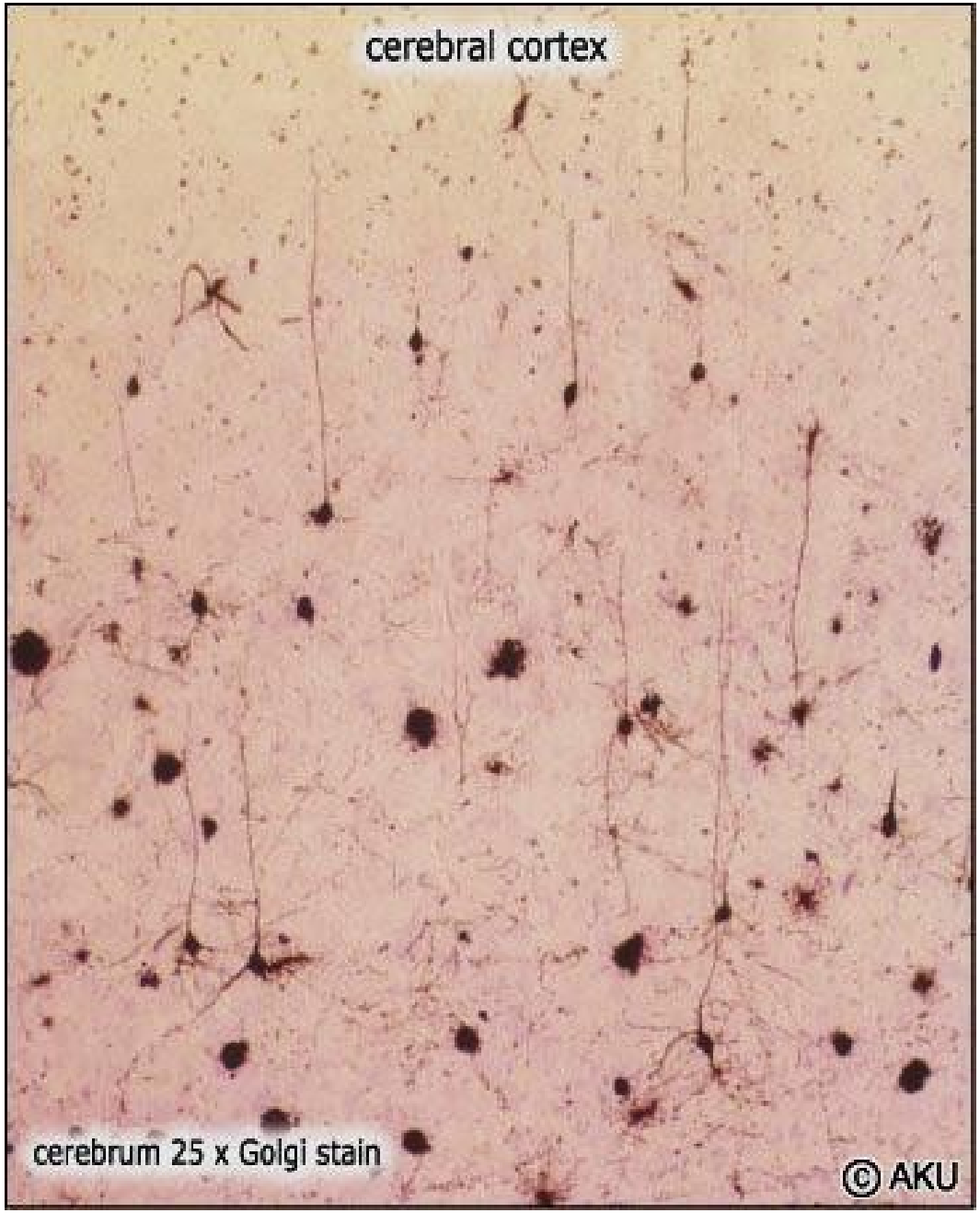
Neurons

- According to no. of processes
  - Unipolar/Pseudo-unipolar
  - Bipolar
  - Multipolar

- According to shape
  - Stellate
  - Pyramidal
  - Fusiform
  - Horizontal

- According to length of processes
  - Golgi Tendon I
  - Golgi Tendon II

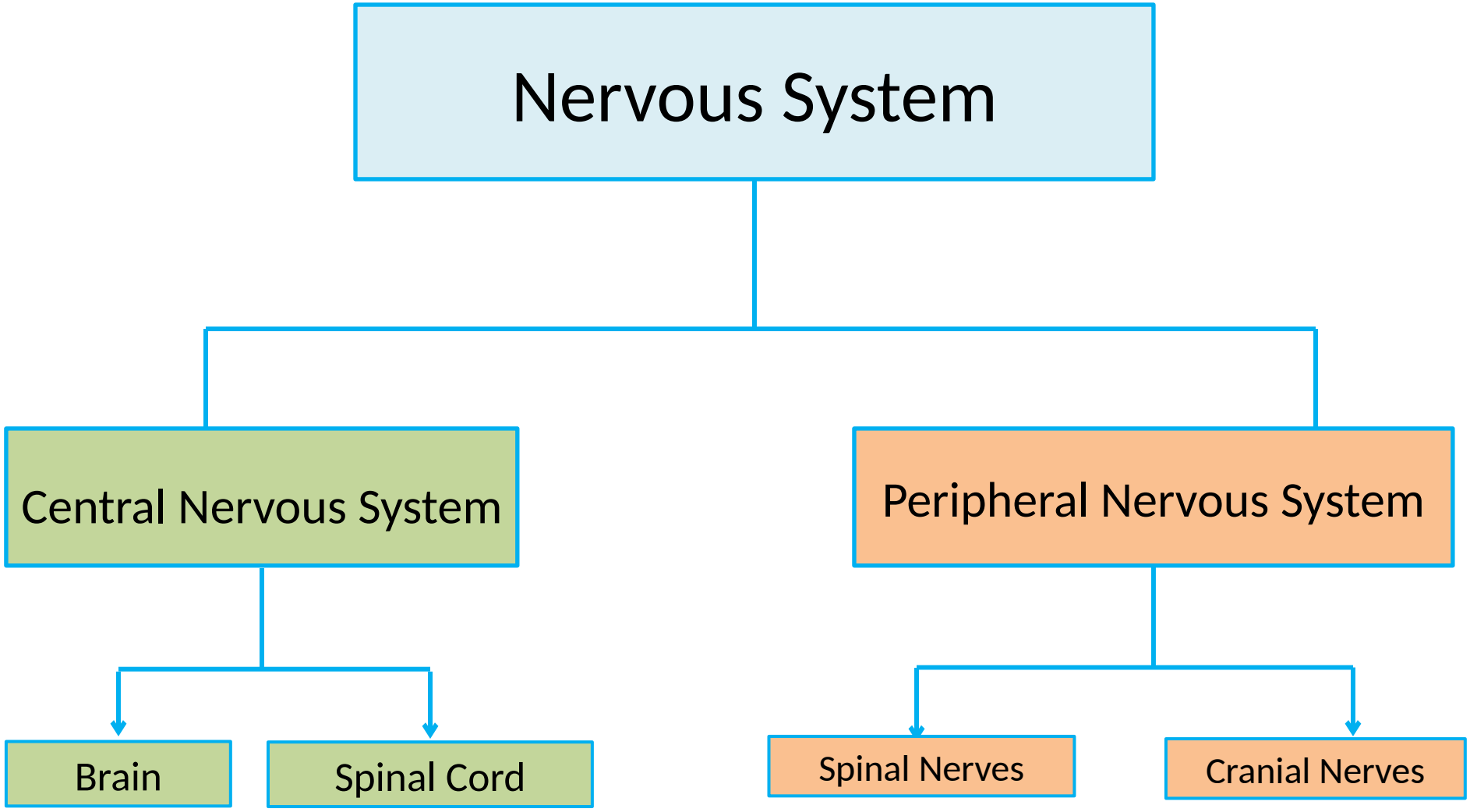
Types of Neurons – According to No. of Processes


cerebral cortex
cerebrum 25 x Golgi stain
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A collection of neurons within the brain or spinal cord (CNS) is called **Nucleus**

A collection of neurons outside the brain & Spinal cord (PNS) is called **Ganglion** *(pl. Ganglia)*

A collection of axons is called as **Tract** inside CNS and **Nerves** when in periphery


Nervous System
Central Nervous System
Peripheral Nervous System
Brain
Spinal Cord
Spinal Nerves
Cranial Nerves

Central Nervous System

Comprises brain and Spinal cord

Specialized tissue with two types of cells

Excitable cells and non excitable cells

Non excitable cells are called **Glial** cells

Peripheral Nervous System

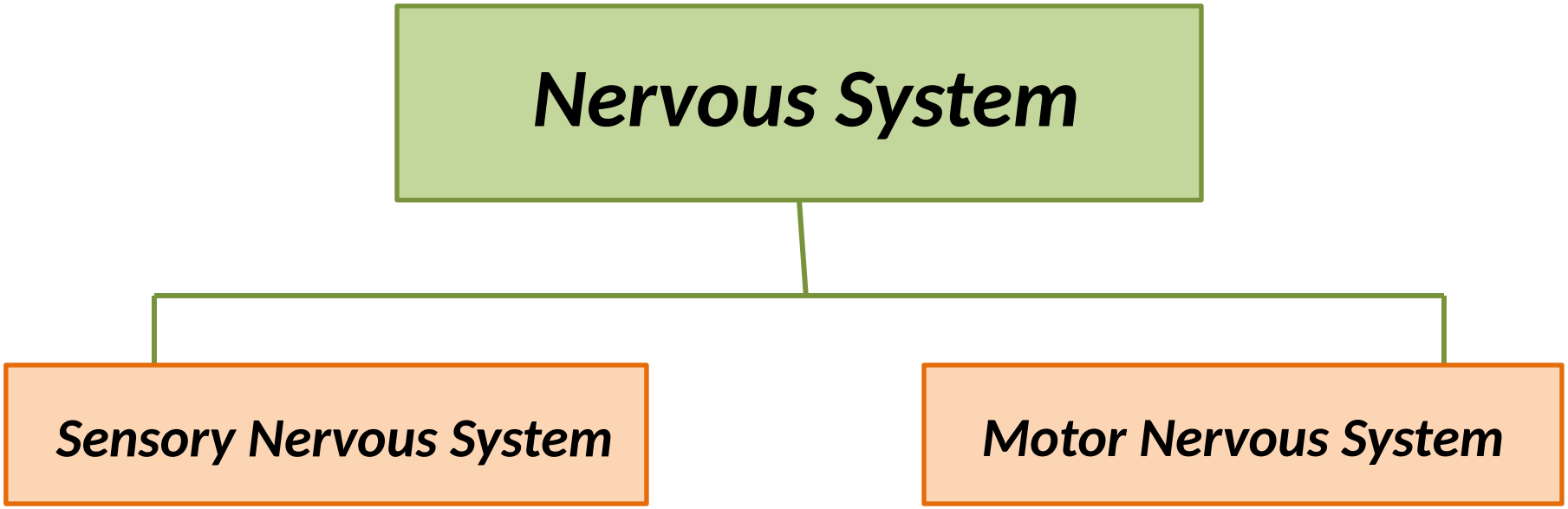
Two components:

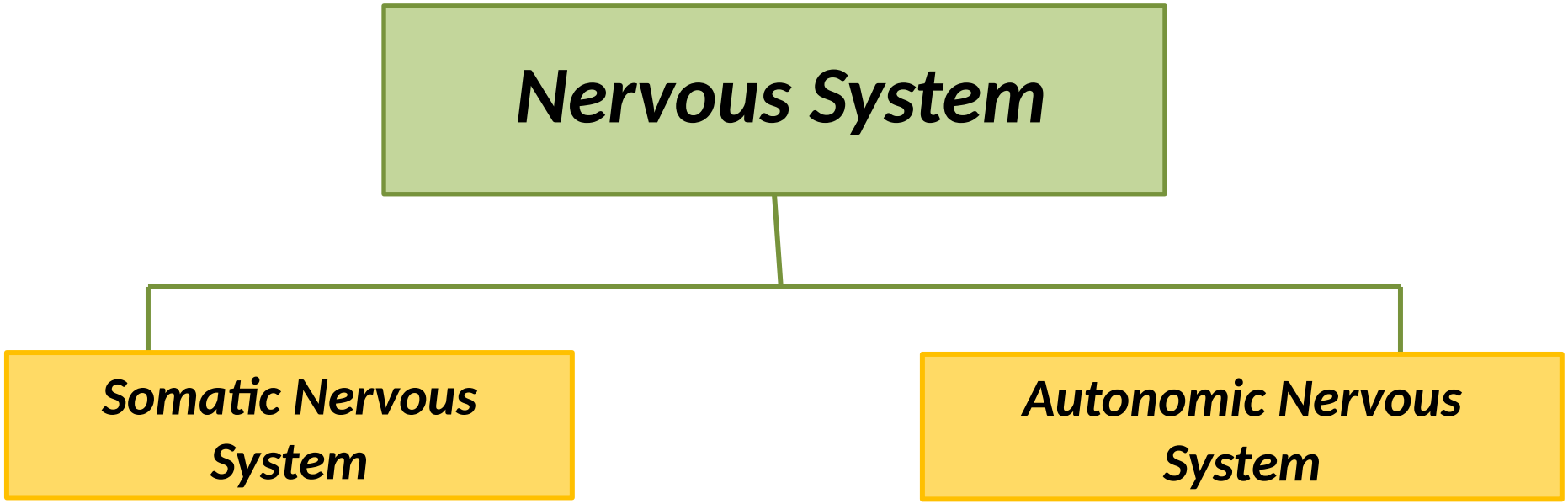
A set of nerves arising from brain and spinal cord

Also, their associated **Ganglia**

Nerve --- Bundle of axons and supporting CT sheath

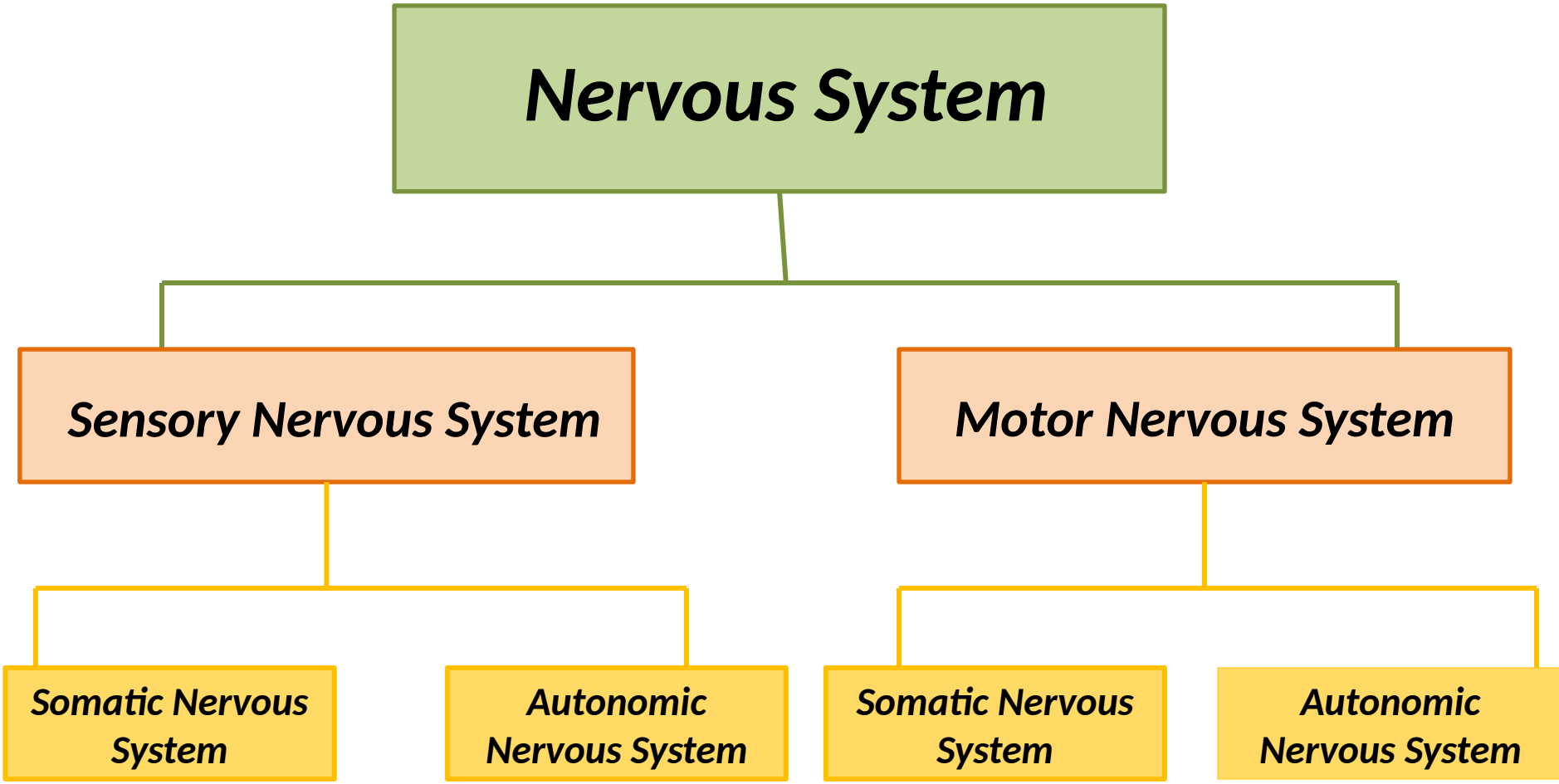
31 pairs of Spinal & **12** pairs of Cranial nerves


Nervous System
Sensory Nervous System
Motor Nervous System

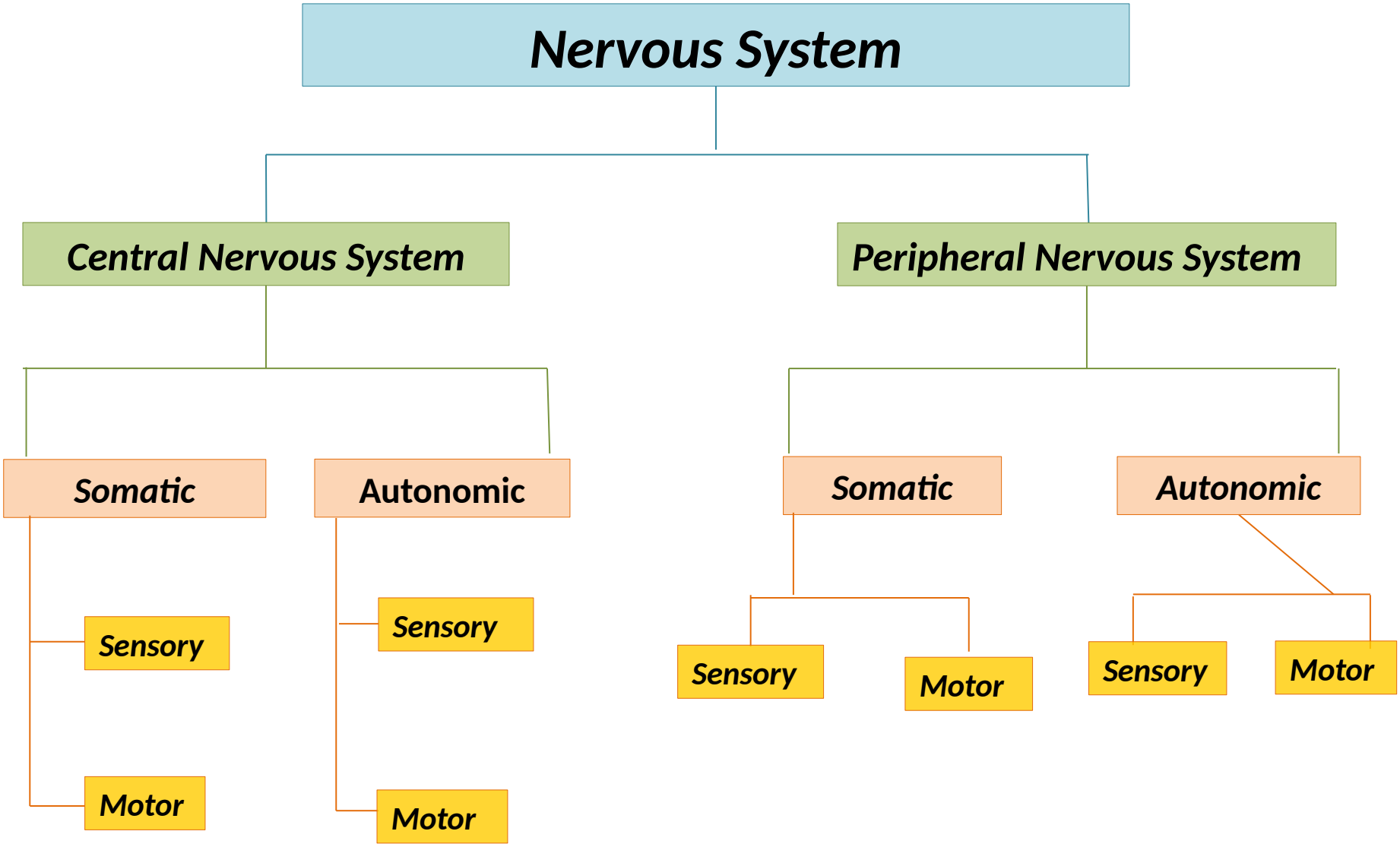

Nervous System
Somatic Nervous System
Autonomic Nervous System

Somatic v/s Autonomic NS

Soma/ body wall	Organs/ Viscera
Skeletal muscles	Smooth & Cardiac m.
Voluntary in action	Involuntary in action
e.g. Axial & Appendicular muscles	e.g. organs of alimentary canal, Intraocular muscles


Nervous System
Sensory Nervous System
Motor Nervous System
Somatic Nervous System
Autonomic Nervous System
Somatic Nervous System
Autonomic Nervous System

Lets Recap!


Nervous System
Central Nervous System
Peripheral Nervous System
Somatic
Autonomic
Somatic
Autonomic
Sensory
Motor
Sensory
Motor
Sensory
Motor
Sensory
Motor

Quiz

- What are the main parts of nervous system?
- What comprises the Central nervous system?
- What comprises the Peripheral n. system?
- How many spinal & cranial nerves?
- Define Nucleus, Ganglion & Tract
- How many types of cell does nervous tissue may have?

Thank you

- References:

  Last's Anatomy 11th edition

  Snell's Clinical Anatomy 5th edition

  Dorland's Dictionary 28th edition

- Courtesy:

  AKU's Digital Atlas of Microscopic Anatomy